CliniScan AI – Detection Module Report

(Milestone 4 Submission: Chest X-ray Object Detection System)

1. Introduction

The CliniScan AI Detection Module identifies and localizes abnormalities in chest X-ray images using deep learning. It enhances interpretability by providing bounding box predictions using Faster R-CNN with a MobileNet backbone.

2. Objectives

• Detect abnormalities

• Localize disease regions

• Handle multiple objects

• Provide visual outputs

• Integrate with system

3. Model Architecture

Model: Faster R-CNN
Backbone: MobileNet
Framework: PyTorch
Classes: 16

Components: RPN, ROI Pooling, Classification Head, Bounding Box Regression

4. Dataset and Preprocessing

Dataset: Chest X-rays with annotations
Split: 80% train, 20% validation

Steps: normalization, augmentation, bounding box formatting

5. Training Configuration

Optimizer: AdamW
Learning Rate: 1e-4
Batch Size: 4
Epochs: 4

6. Evaluation Metrics

Precision, Recall, F1 Score, IoU (0.5 threshold)

7. Prediction Pipeline

Input image → Region proposals → NMS + threshold → Final bounding boxes

8. Visualization

Ground truth: green dashed boxes
Predictions: red solid boxes

9. Performance

Strengths: Good single-object detection
Limitations: Difficulty with multiple overlapping objects

10. Challenges

Multiple detection handling, overlapping boxes, dataset imbalance

11. Improvements

Applied NMS, threshold tuning, improved visualization

12. Future Work

Soft-NMS, larger dataset, YOLOv8 exploration

13. Backend API

Flask backend handles requests, runs inference, and returns results.

Endpoints:
- /api/analyze (POST)
- /api/stats (GET)
- /api/history (GET)

14. Deployment

Frontend → Backend → Model → Output

15. Conclusion

The system successfully detects and localizes abnormalities in chest X-rays.

16. Website Screenshots

The following screenshots illustrate the Lung Assist web application built for the CliniScan AI system, developed by Humane Diagnostics.

Figure 1: Home Page – AI-Powered Chest Analysis Overview

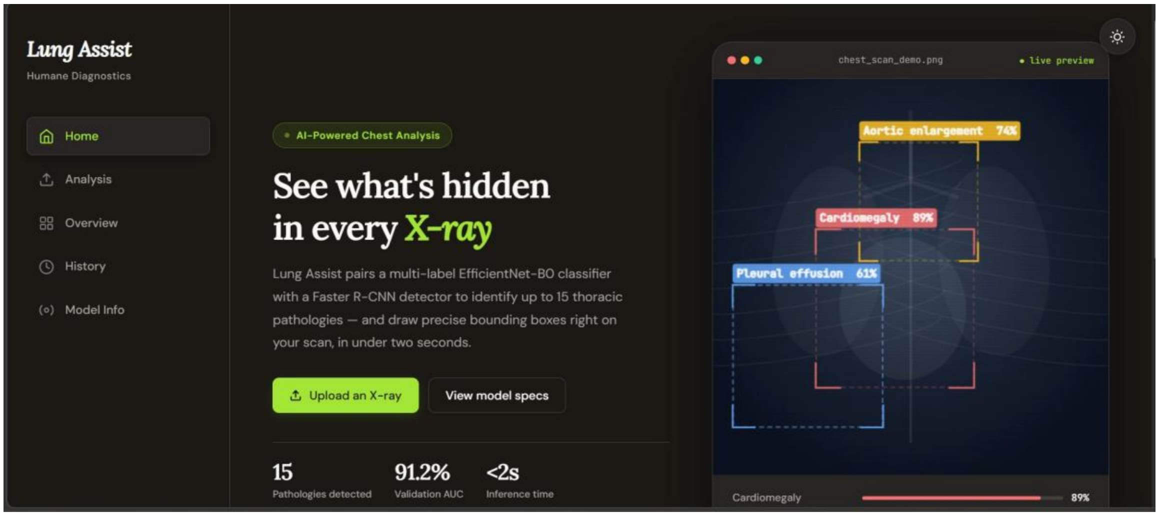


Figure 2: Home Page – Features and How It Works

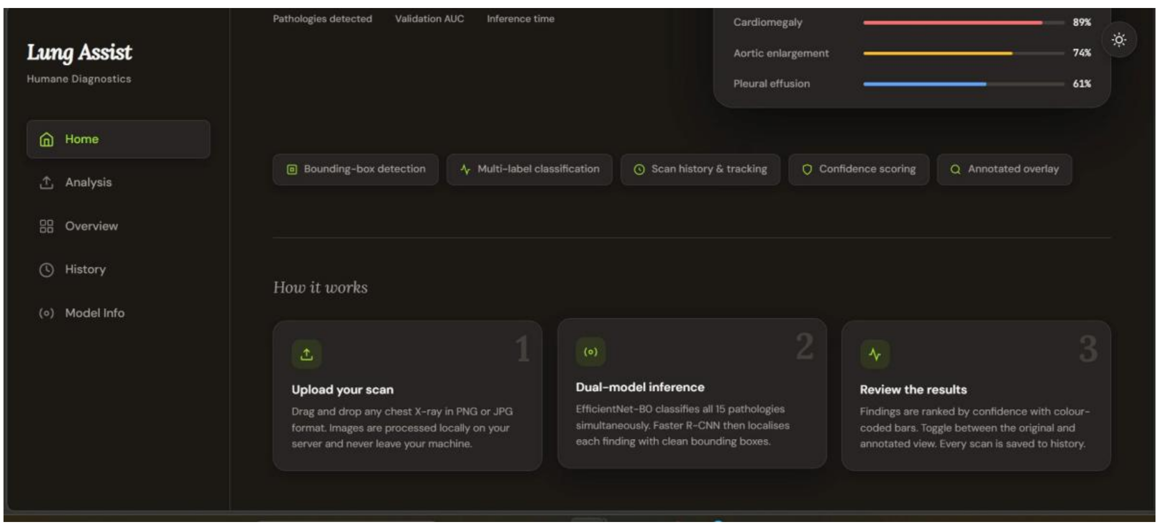


Figure 3: Analysis Page – Upload Interface

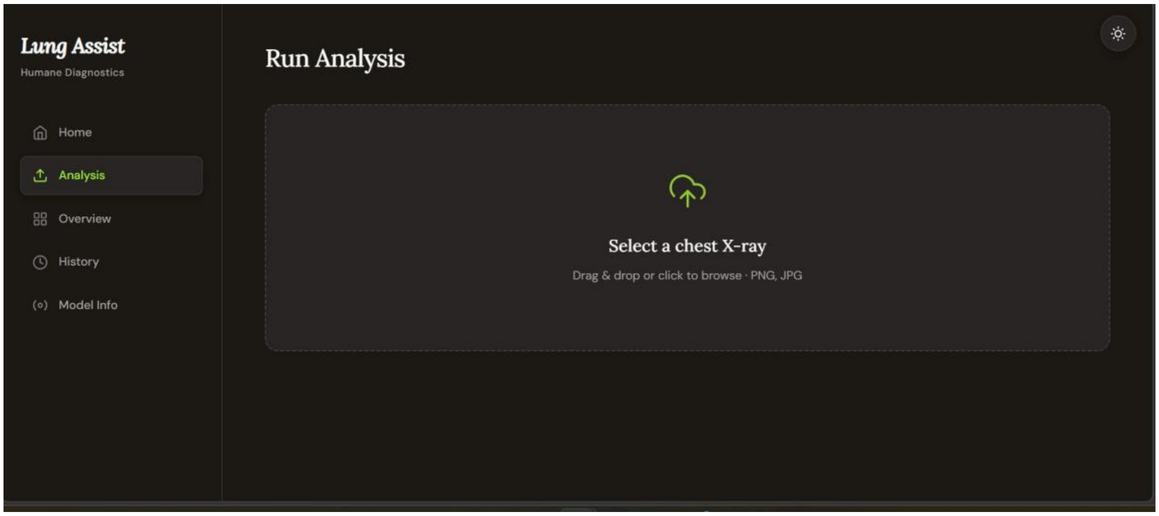


Figure 4: Overview Page – Scan Statistics Dashboard

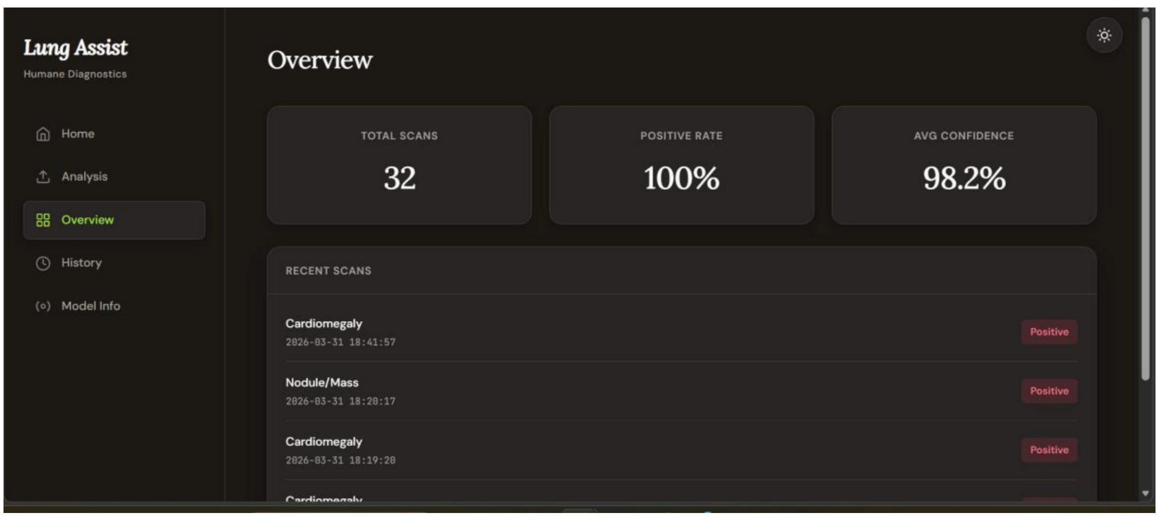


Figure 5: Analysis Results – Annotated X-ray with Findings

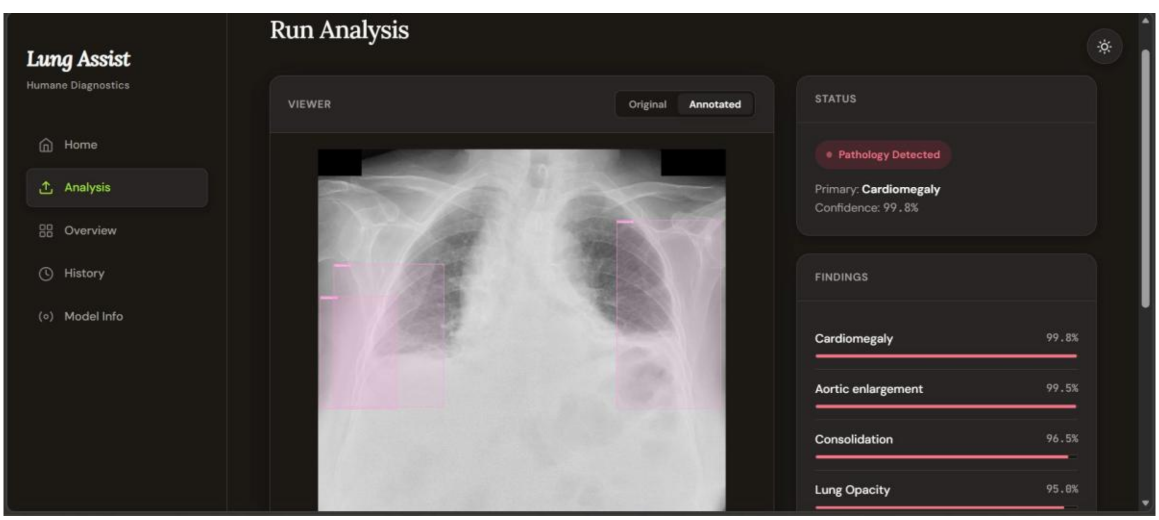


Figure 6: Scan History Page

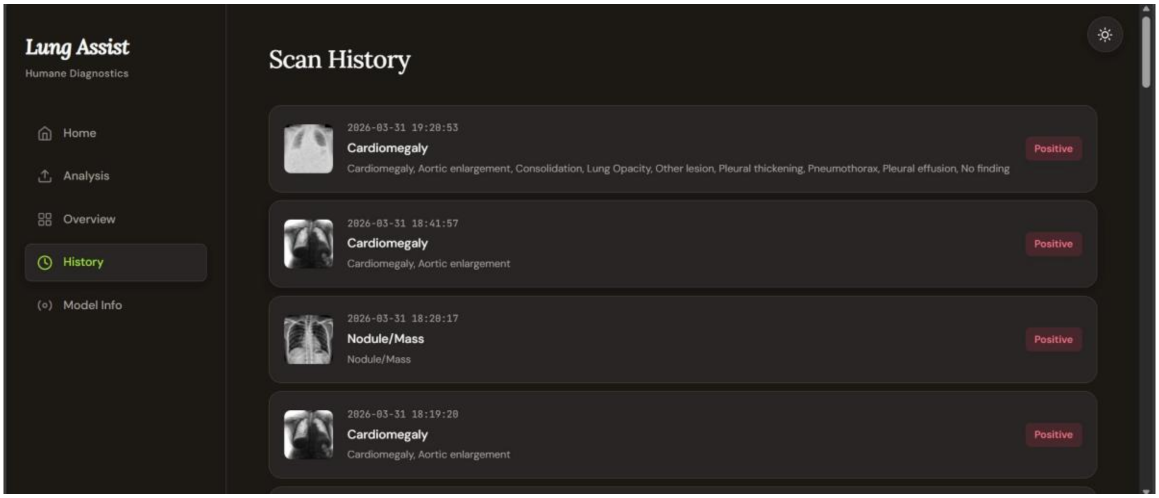


Figure 7: Model Info Page – Classifier and Detector Specifications

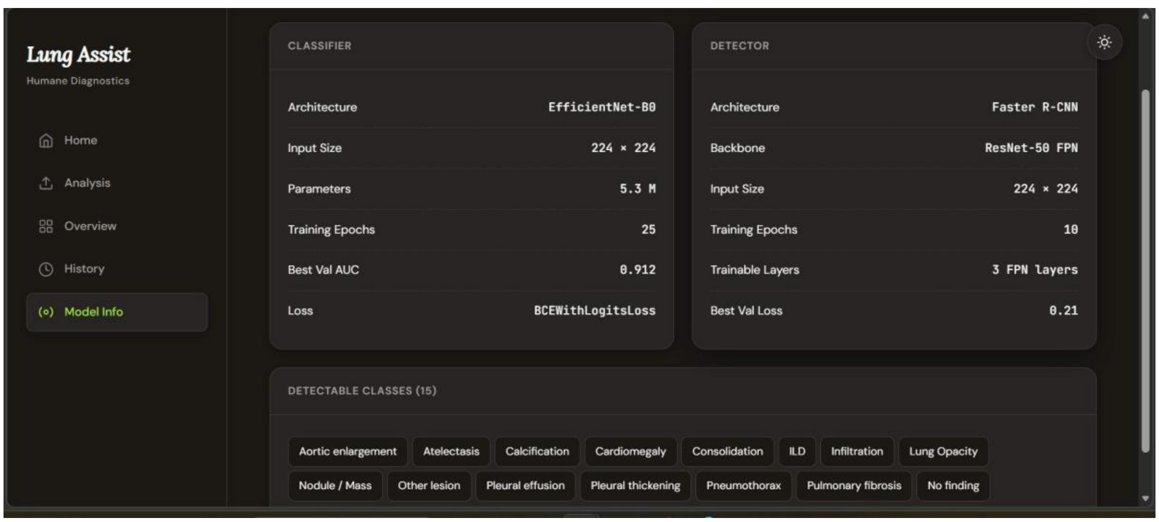
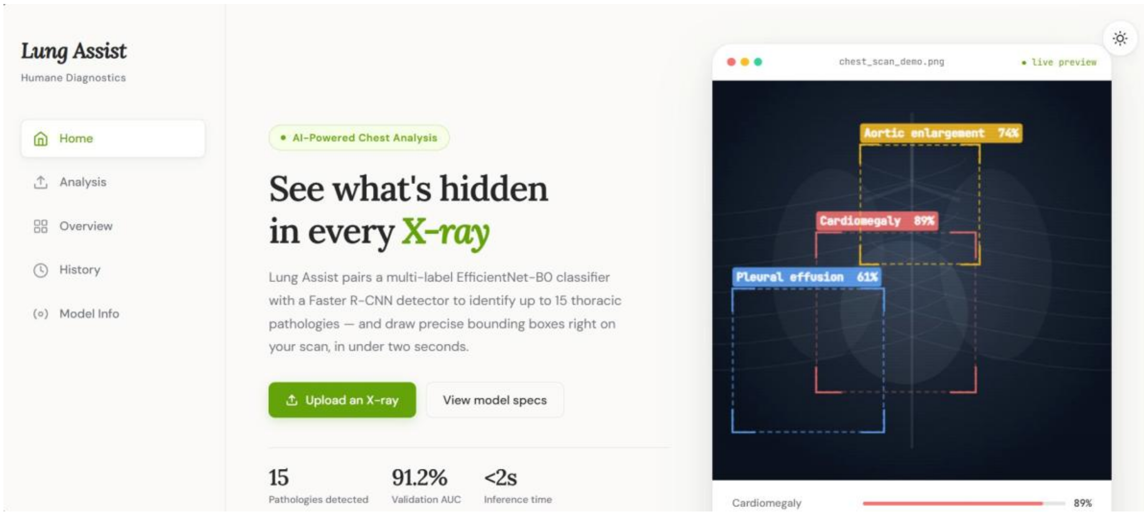


Figure 8: Home Page – Light Mode View



Project Link

https://huggingface.co/spaces/poojasri06/LungAbnormality